COMP8325 Applications of Artificial Intelligence for Cyber Security

Group Project Description
Total Marks: 30
Weighting: 30%

MACQUARIE
University

DEADLINE:

- Group Formation Due: Friday, 24 April, 11:55 PM (End of Week 7).
- Source Code + Report + Presentation Slides: Friday, 29 May, 11:55 PM (End of Week 12).
- Presentation: In person, Monday, 1 June, 9:00 AM to 11:00 AM (Week 13 Lecture).

LEARNING OUTCOME

By completing this assignment, you should demonstrate your ability to:

- Understand and detect abnormal patterns in a variety of real-world datasets for cyber security
- Perform data pre-processing, data exploration, and feature selection on various types and volumes of data for different cyber security applications
- Perform machine learning training and evaluation for cyber security applications
- Understand the security threats to machine learning systems deployed for cyber security
- Communicate professionally in written and oral form to a range of audiences

SUBMISSION

This is a group project. Each group consists of 4 to 5 students. Submission is required per group. Each group needs to submit three files:

(1) Source code: Three Jupyter Notebook files (i.e., with the extension, `.ipynb`) with the implementation and analysis of the Data Analysis Task.
   - See Section 1.1 and Section 1.2 for details of Task 1 and Task 2, respectively.
   - Since this is a collaborative project, you can use tools such as Github which allows your team to make collaborative edits. See for example: https://reproducible-science-curriculum.github.io/sharing-RR-Jupyter/01-sharing-github/

(2) Report: A PDF file for Task 1 (Model Training) and Task 2 (Evaluation on Holdout Dataset).

- See Section 1.1 and Section 1.2 for details of Task 1 and Task 2, respectively.
- The report must be between 3 and 5 pages (excluding references, if applicable). A separate title page is not required.
- Student details (name and student ID) must be clearly stated at the beginning of the report, with one line per student.

(3) Presentation slides: A PDF version of the slides to be presented for Task 3.
- The slides themselves will be presented in person by all group members.
- See Section 1.3 for details of Task 3.

All submissions are through iLearn via Turnitin submission links. At the beginning of each file, please write down your group name.

MARKING CRITERIA

Tables 1 and 2 present comprehensive marking rubrics for the Data Analysis and Research tracks of the group project, respectively. A summary of the key evaluation criteria is provided below:

- Clarity of organisation, presentation, and readability of the Jupyter notebooks and final report.
- Quality, structure, and readability of the source code.
- Demonstration of critical thinking and depth of understanding in interpreting the results of the data analysis.
- Depth and quality of the literature review, including a clear understanding of related work.
- Effectiveness of the team's presentation, reflecting professional communication skills and coordination.

LATE SUBMISSIONS

- Late submissions follow the policies specified in the Unit Guideline.
- If you have a legitimate reason for submitting late, discuss this with the convenor before the assignment due date.

1. Project Details

Your main task is to train **two machine learning classifiers** of your choosing to predict malware categories using the **BODMAS malware dataset**[1]. BODMAS is short for Blue Hexagon Open Dataset for Malware Analysis. This is a labelled dataset containing labels for different categories of malware, including a separate category for benign programs. This dataset, available at Google Drive[2] contains three files:

(1) `bodmas.npz`: this contains feature vectors extracted from the programs and a label which is either 0 for benign or 1 for malware. These features are extracted from the LIEF project.
(2) `bodmas_metadata.csv`: this contains the SHA-256 hashes of programs and their fine-grained malware families. The benign programs are under the family `NaN`.
(3) `bodmas_malware_category.csv`: this contains the SHA-256 hashes of programs and their malware categories. This is the label that you need to predict. This file does not contain any benign programs.

Note that none of the datasets contain actual benign and malware programs, only the features and labels extracted from them.

DATA HANDLING

You must read the following instructions before processing the dataset.

[1] https://whyisyoung.github.io/BODMAS/
[2] https://drive.google.com/drive/folders/1Uf-LebLWyi9eCv97iBal7kL1NgiGEsv_?usp=drive_link

- You will need to preprocess the datasets to extract a dataset containing feature vectors with the corresponding malware category as their label.
- The ith feature vector in `bodmas.npz` corresponds to the ith row in `bodmas_metadata.csv`.
- The malware category labels are in `bodmas_malware_category.csv`. In particular, do not use the malware family names in `bodmas_metadata.csv` as your labels. See the instructions in https://whyisyoung.github.io/BODMAS/ on how to read the `bodmas.npz` dataset.
- You should only use the dataset uploaded to the link mentioned above and not the one on the website.
- Further details about the features and their meanings can be found here: https://github.com/elastic/ember

1.1. **Task 1: Model Training (18 marks).** A report and source code on the performance of the two machine learning algorithms in classifying malware categories. The report should include details of

- Breakdown of roles within the team
- The two machine learning algorithms chosen by your team
- Data preprocessing
- How training and testing were performed, including the train–test split
- How hyperparameters were determined for your classifiers
- Evaluation of training and testing using performance metrics, including:
  - Overall accuracy
  - True Positive Rate (TPR) for each malware category (including benign)
  - False Positive Rate (FPR) for each malware category (including benign)
- A descriptive comparison of how the two algorithms perform against each other

Deliverables

- A Jupyter notebook (`.ipynb`) containing the implementation
- A PDF report summarising the methodology and results

1.2. **Task 2: Evaluation on Holdout Dataset (4.5 marks).** Within one week before the final submission, a **holdout dataset** will be released on iLearn. Your task is to run your **already trained classifiers** on this holdout dataset and report:

- Overall accuracy
- True Positive Rate (TPR) for each malware category (including benign)
- False Positive Rate (FPR) for each malware category (including benign)

Important note: You are **not allowed to retrain** the model after the release of the holdout dataset. The holdout dataset will be formatted identically to the dataset used for training.

Deliverables

- Updated Jupyter notebook results on the holdout dataset
- Updated report section discussing holdout performance

1.3. **Task 3: Presentation (7.5 marks).** You are required to deliver a **five-minute in-class presentation** covering:

- The machine learning algorithms used by your team
- How hyperparameters were selected
- Performance on the original dataset
- Performance on the holdout dataset and reflection on any performance differences

Notes:

- The presentation time is strictly **5 minutes**.
- Do not prepare more than **5 slides**, including the title slide.
- Marks are awarded based on presentation quality and clear explanation of results, not on achieving the best classifier performance.

Deliverables

- A PDF version of the presentation slides
- A 5-minute in-class presentation

Table 1. Marking rubric for Task 1: Model Training.

Marks	**Marking Rubric for Task 1**
1.5	Accomplishment of all required tasks
1.5	Well-organised, presented, and readable (Jupyter notebook and report)
3.0	Discussion and justification of the selected machine learning algorithms
1.5	Quality, readability, and reproducibility of source code
2.0	Data loading and label preparation
3.0	Data preprocessing and feature handling
2.5	Training methodology, train–test split, and hyperparameter tuning
3.0	Evaluation using appropriate metrics, including overall accuracy, TPR, and FPR for each class
18	**Subtotal**

Table 2. Marking rubric for Tasks 2 and 3.

Marks	**Marking Rubric for Task 2: Holdout Dataset Evaluation**
1.0	Correct use of the already trained classifiers on the holdout dataset
1.0	Clear reporting of overall accuracy on the holdout dataset
1.0	Correct reporting of TPR and FPR for each malware category (including benign)
1.5	Clear interpretation and reflection on holdout performance
4.5	**Subtotal**
Marks	**Marking Rubric for Task 3: Presentation**
2.0	Clear overview of the machine learning algorithms used
1.5	Explanation of hyperparameter selection
1.5	Clear presentation of results on the original dataset
1.5	Clear presentation of results on the holdout dataset and reflection on differences
1.0	Professional communication, slide quality, and delivery
7.5	**Subtotal**
30	**Grand total**